

Zhengwei Zhang

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Updated on June 10, 2026

Education

- 2021–Present **Ph.D. in Quantitative Marketing**, The University of Texas at Austin
- 2018–2019 **M.S. in Finance**, Johns Hopkins University
- 2017–2018 **M.S. in Finance**, Lancaster University
- 2013–2017 **B.M.S. (Major in Accounting)**, Zhongnan University of Economics and Law

Research Interests

- Methodological** Bayesian Inference, Field Experiments
- Substantive** Electric Vehicles, AI Policy Design

Job Market Paper

Capturing the Impact of Tesla: Dynamic Factor Analysis Using Online Search, Advertising and Sales

Committee Jason Duan (chair), Garrett Sonnier (member), Alain Lemaire(member), Robert Peterson (external member), Yuling Yao (external member)

Status Preparing for Submission to *Marketing Science*

Abstract This paper examines how the introduction of the Tesla Model 3 reshaped consumer attention in the U.S. automobile market. Prior research shows that online search is an important intermediate stage in the purchase funnel, especially for high-involvement durable goods. However, less is known about how search interest moves dynamically between broad category-level topics and specific products. Using a weekly vehicle-level dataset from 2014 to 2021 that combines Google Trends search indices, Vivvix advertising expenditures, and Ward's vehicle sales, I study how online search moves between broad electric-vehicle topic interest and model-specific vehicle interest. I develop a dynamic latent-factor model that treats observed search measures as noisy indicators of underlying consumer information states and allows these states to spill over across the EV topic, Tesla Model 3, non-Tesla electric vehicles, and incumbent compact competitors. The results show that the Model 3's primary impact appears in the search system rather than in immediate sales or advertising responses. After the Model 3 entry, EV-topic attention became more persistent, Tesla Model 3 search became a major driver of broader EV interest, and the competitive search environment became more organized around EV-centered comparison paths. These findings suggest that disruptive product entry can reshape the structure of consumer attention before producing detectable sales effects.

Works in Progress

When Safety Constrains Creativity: Evidence of AI Content Moderation on User Engagement (with Runyang Shu and Yixing Chen)

As generative AI platforms scale and face heightened legal, copyright, and brand-safety scrutiny, providers are increasingly tightening content moderation policies. While stronger guardrails can improve compliance, platform legitimacy, and risk management, they may also reduce perceived creative latitude and raise refusal rates—potentially shaping user engagement, retention, and subscription demand. In partnership with a large technology firm operating a commercial AI content-generation platform, we examine the behavioral and economic effects of content moderation tightening, quantifying how shifts in policy strictness translate into changes in user activity and monetization outcomes.

Linear and Nonlinear Dynamic Factor Models for Spillover Effects and Forecasting of Electric Vehicle Diffusion(with Jason Duan and Garrett Sonnier)

As electric vehicles reshape the automotive market, consumer attention, marketing activity, and sales outcomes are increasingly intertwined across EV and traditional segments. While sales data alone provides limited visibility into upstream demand formation, multidimensional online search behavior (e.g., web, image, and YouTube queries) offers a richer lens on discovery, evaluation, and purchase intent—yet it exhibits strong comovement, nonlinearity, and outliers that standard linear time-series models often miss. I study these correlated dynamics by integrating Google Trends, advertising expenditures, and vehicle sales to quantify category and product spillovers and to improve forecasting of EV diffusion. Building on linear dynamic factor models for interpretable measurement of shared demand forces, I develop a nonlinear extension that retains a factor-structured observation equation while using neural network state dynamics to flexibly capture regime shifts and nonlinear responses, enabling more accurate and behaviorally grounded inference on spillover effects and adoption trajectories.

Work Experience

- 2026 **Senior Data Analyst**, *Wondershare*
Worked on designing AI policy for content management
- 2026 **Data Analyst**, *Vmax New Energy*
Provided market analysis for the electric control systems section

Teaching Experience

- Spring 2024 **Instructor**, *The University of Texas at Austin*, Principles of Marketing
Overall Instructor Rating: 4.5/5

Conference Presentations & Attendance

- 2022 AMA Sheth Foundation Doctoral Consortium
- 2024 UH Marketing Doctoral Symposium
- 2024 Bass FORMS Conference
- 2024 Marketing Science Conference
- 2024 ISMS Doctoral Consortium
- 2024 Quantitative Marketing and Economics Conference
- 2025 Marketing Dynamics Conference

Honors, Awards & Grants

- 2021–2026 Recruitment Graduate School College Fellowship
- 2021–2026 McCombs School Fellowship
- 2023 Bonham Eugene and Dora Memorial Fellowship

Skills

- Programming R; Python; MATLAB; Stan
- Languages English (proficient); Mandarin (native); Cantonese (conversational)

■■■■■ Dissertation Committee

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